

Exact Bloch Oscillations and Schatten Boundaries for the Wannier–Stark Lattice

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Abstract

For the uniform-field tight-binding Hamiltonian on $\ell^2(\mathbb{Z})$ we give one all-parameter closure: a periodic Fourier gauge, the simple Wannier–Stark ladder and Bessel basis, the exact propagator and least identity return. We also derive the delta-source probability shell and second moment, including the zero-hopping and changed-owner zero-field faces. No infinite-dimensional conclusion is inferred from finite truncation.

1 Frozen owner and unitary normal form

For $F, J \in \mathbb{R}$, $F \neq 0$, set

$$(H\psi)_n = Fn\psi_n - J(\psi_{n+1} + \psi_{n-1}) \quad (1)$$

on the domain of multiplication by n in $\ell^2(\mathbb{Z})$. This is the one-band uniform-field lattice associated with the Wannier–Stark problem [1]; exact lattice propagators in this setting were studied by Yellin [2]. These citations establish lineage, not novelty.

Fix the convention $(\mathcal{F}\psi)(k) = \sum_{n \in \mathbb{Z}} \psi_n e^{ink}$ into $L^2(\mathbb{T}, dk/2\pi)$. Then

$$\hat{H} = -iF\partial_k - 2J \cos k.$$

Let G multiply by $\exp[-i(2J/F) \sin k]$. Direct differentiation gives

$$\hat{H} = G^{-1}(-iF\partial_k)G. \quad (2)$$

The sign in (2) is part of the contract: reversing it reverses the hopping.

Theorem 1 (ladder and basis). *H has simple pure-point spectrum $F\mathbb{Z}$. A complete orthonormal eigenbasis is*

$$\phi_m(n) = J_{n-m}(2J/F), \quad H\phi_m = Fm\phi_m. \quad (3)$$

Proof. The circle momentum in (2) has the simple orthonormal eigenbasis e^{imk} . Applying G^{-1} and using $e^{ia \sin k} = \sum_r J_r(a) e^{irk}$ gives (3), including its index. Equivalently, the Bessel recurrence $J_{r-1}(a) + J_{r+1}(a) = 2rJ_r(a)/a$ verifies the eigen-equation. Gauge unitarity supplies completeness and orthonormality, rather than a finite-volume limit. $\square$

2 Exact evolution and least return

Theorem 2 (propagator). *For all $m, n \in \mathbb{Z}$ and $t \in \mathbb{R}$,*

$$U_{nm}(t) = i^{n-m} e^{-iFt(n+m)/2} J_{n-m}\left(\frac{4J}{F} \sin \frac{Ft}{2}\right). \quad (4)$$

The least positive T for which $U(T) = I$ on the full space is $T_B = 2\pi/|F|$.

Proof. Solving the Fourier transport equation along characteristics, or inserting (3) and applying the Bessel addition formula, yields (4). Its off-diagonal sentinel is $U_{m+1,m}(t) = +iJt + O(t^2)$, as required by $U(t) = I - itH + O(t^2)$. At T_B , the Bessel argument is zero and the diagonal phase is one, so $U(T_B) = I$. Conversely, $U(T) = I$ applied to the eigenvalues 0 and F forces $e^{-iFT} = 1$; hence no smaller positive return exists. $\square$

3 Probability shell, moment, and boundary atlas

For $\psi(0) = \delta_0$, put $z(t) = (4J/F) \sin(Ft/2)$. Formula (4) gives the exact shell

$$\mathbb{P}\{X_t = n\} = |U_{n0}(t)|^2 = J_n(z(t))^2. \quad (5)$$

The characteristic function of (5) is $J_0(2z \sin(q/2))$. Its value and first two derivatives at zero give

$$\sum_n \mathbb{P}\{X_t = n\} = 1, \quad \mathbb{E}X_t = 0, \quad \mathbb{E}X_t^2 = \frac{z(t)^2}{2} = \frac{8J^2}{F^2} \sin^2 \frac{Ft}{2}. \quad (6)$$

Thus spreading is bounded and returns exactly, not just asymptotically.

When $J = 0$, (1) is diagonal: (2)–(4) and the least return remain valid, while (5) is frozen at zero. The limit $F = 0$ with $J \neq 0$ changes owner. Fourier multiplication by $-2J \cos k$ has absolutely continuous spectrum $[-2|J|, 2|J|]$, kernel $i^{n-m} J_{n-m}(2Jt)$, and no positive full-space identity return. It is therefore not inserted into the $F \neq 0$ theorem.

References

- [1] G. H. Wannier, *Wave Functions and Effective Hamiltonian for Bloch Electrons in an Electric Field*, Phys. Rev. **117** (1960), 432, [doi:10.1103/PhysRev.117.432](https://doi.org/10.1103/PhysRev.117.432).
- [2] J. Yellin, *Two exact lattice propagators*, Phys. Rev. E **52** (1995), 2208, [doi:10.1103/PhysRevE.52.2208](https://doi.org/10.1103/PhysRevE.52.2208).